MILITARY SPECIFICATION

# FUZE, ROCKET, BASE DETONATING, XM434E1 ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-F-50220A (MU), dated 19 June 1970.

Page 1

Add the following Preamble underneath the title: "This specification is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense."

PAGE 5

4.2.1, Line 4: Delete "40 complete fuzes," and substitute "60 complete fuzes".

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4.2.1, Last sentence: Delete and substitute the following: "A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government."

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4.3.3.1.7, Line 2: Delete "40 fuzes," and substitute "60 fuzes".

4.3.3.1.7.1: Delete the first 3 lines and substitute the following: "Jolt and Jumble (see 3.9). Failure to withstand test - Critical Defect- Code No. 16012; Missing plate screws - Major Defect - Code No. 16007. - Ten of the 60 fuzes selected in 4.3.3.1.7 "

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4.3.3.1.7.2: Delete in its entirety and substitute the following: "Rough Handling (see 3.10). - The remaining 50 fuzes shall be subjected to the transportation-vibration test and then without disassembly, to the five-foot drop test specified in 3.10."

4.3.3.1.7.2.1: "Arming Tests. - These tests shall be performed on thirty (30) of the fifty (50) fuzes only. The cover assembly, graze elements and base plug components shall be removed but the graze elements (firing pin and firing pin spring) identified for later re-insertion into the same fuze and the fuzes shall be subjected to the following tests in the sequence specified using test equipment specified in 4.3.4.

| TEST | TEST PROCEDURE |
|---|---|
| Non-functioning of setback assembly | 4.4.8 |
| Functioning time of setback assembly | 4.4.8 |
| Rotor arming time at plus 70 degrees F | 4.4.9 |
| Rotor arming time at minus 40 degrees F | 4.4.9 |

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70 degrees F and minus 40 degrees F shall be plus and minus 5 degrees F.

The first article sample shall be rejected if one or more defectives are found according to the defects listed in Table 1 below. The results may be examined and tested for outliers, but only one outlier is allowed and only for the minus 40 degree F temperature. Refer to 6.6 and 6.7 for calculations of sample standard deviation (sigma) and outliers respectively. After completion of testing (see 4.4.9), the cover assemblies, graze elements and base plugs shall be reassembled to the fuzes and the fuzes tested as follows in 4.3.3.1.7.2.2."

TABLE 1

ARMING TEST DEFECTIVES

| Defect | Category | Code No. |
|---|---|---|
| 1. Functioning during non-functioning test of setback assembly (see 3.6.1) | Critical | 16008 |
| 2. Functioning time of setback assembly exceeds maximum (see 3.6.2) | Major | 16009 |



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|---|---|---|
| 3. Rotor arming time @ 70 degrees F less than minimum (see 3.8.1) | Critical | 16010 |
| 4. Rotor arming time @ 70 degrees F more than maximum (see 3.8.1) | Major | 16011 |
| 5. Rotor arming time @ minus 40 degrees F less than minimum (see 3.8.2) | Critical | 16015 |
| 6. Rotor arming time @ minus 40 degrees F more than maximum (see 3.8.2) | Major | 16016 |
| 7. Rotor fails to arm @ either temperature | Major | 16017 |
| 8. Average rotor arming time plus 1.90 sigma exceeds maximum specified time @ either temperature | Major | 16018 |
| 9. Average rotor arming time minus 1.90 sigma is less than the minimum time specified @ either temperature | Major | 16019 |

4.3.3.1.7.2.2: "Explosive Graze Functioning (see 3.7). - The fifty (50) fuzes previously tested in 4.3.3.1.7.2 and 4.3.3.1.7.2.1 shall be subjected to the test specified in 4.4.10 using test equipment specified in 4.3.4. The first article sample shall be rejected if one or more fuzes are classed defective according to the defects defined in Table II. Fuzes which do not function shall not be included in the calculations for average graze functioning level and outliers are not permitted. Refer to 6.6 for calculation of sample standard deviation (sigma).

TABLE II

EXPLOSIVE GRAZE FUNCTIONING TEST DEFECTIVES

| Defect | Category | Code No. |
|---|---|---|
| 1. Fuze fails to arm @ maximum acceleration level (see 3.6.2) | Major | 16017 |
| 2. Fuze functions during non-functioning graze acceleration test (see 3.7) | Major | 16013 |



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|---|---|---|
| 3. Fuze fails to function explosively or explosive elements fail to perforate the booster cup | Major | 16020 |
| 4. Average graze functioning level plus or minus 1.60 sigma fails to satisfy the specified requirements (see 3.7) | Major | 16014 |

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4.3.3.1.7.3: Delete in its entirety.

PAGE 19

4.3.3.2.7, Lines 1 and 2: Delete "A sample of 40 fuzes", and substitute "A sample of 60 fuzes".

4.3.3.2.7.1, Line 3: Delete "Ten of the 40 fuzes", and substitute "Ten of the 60 fuzes".

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4.3.3.2.7.2: Delete in its entirety and substitute the following: "Rough Handling (see 3.10). - The remaining 50 fuzes shall be subjected to the transportation-vibration test and then without disassembly to the five-foot drop test specified in 3.10."

4.3.3.2.7.2.1, "Arming Tests. - These tests shall be performed on thirty (30) of the fifty (50) fuzes only. The cover assembly, graze elements and base plug components shall be removed but the graze elements (firing pin and firing pin spring) identified for later re-insertion into the same fuze and the fuzes shall be subjected to the following tests in the sequence specified using test equipment specified in 4.3.4.

| TEST | TEST PROCEDURE |
|---|---|
| Non-functioning of setback assembly | 4.4.8 |
| Functioning time of setback assembly | 4.4.8 |
| Rotor arming time at plus 70 degrees F | 4.4.9 |
| Rotor arming time at minus 40 degrees F | 4.4.9 |

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70 degrees F and minus 40 degrees F shall be plus and minus 5 degrees F.



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The lot shall be rejected if one or more defectives are found according to the defects listed in Table 1 (see 4.3.3.1.7.2.1). The results may be examined and tested for outliers, but only one outlier is allowed and only for the minus 40 degree F temperature. Refer to 6.6 and 6.7 for calculations of sample standard deviation (sigma) and outliers respectively. After completion of testing (see 4.4.9), the cover assemblies, graze elements and base plugs shall be reassembled to the fuzes and the fuzes tested as follows in 4.3.3.2.7.2.2."

4.3.3.2.7.2.2, "Explosive Graze functioning (see 3.7). - The fifty (50) fuzes previously tested in 4.3.3.2.7.2 and 4.3.3.2.7.2.1 shall be subjected to the test specified in 4.4.10 using test equipment specified in 4.3.4. The lot shall be

rejected if two (2) or more fuzes are classed defective according to the defects defined in Table 11 (see 4.3.3.1.7.2.2). Fuzes which do not function shall not be included in the calculations for average graze functioning level and outliers are not permitted. Refer to 6.6 for calculation of sample standard deviation (sigma)."

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4.3.3.2.7.3: Delete in its entirety.

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4.4.10, Delete in its entirety and substitute the following: "Explosive Graze Non-Functioning and Functioning. - The assembly shall be placed in the test fixture and subjected to the required arming acceleration. The assembly shall then be observed for proper arming (failures shall be removed without further testing) and reoriented to permit performance of graze testing. The armed assembly shall then be subjected to the minimum non-functioning graze acceleration specified for the minimum time period specified. If functioning does not occur; the acceleration shall be increased until functioning occurs and the fuctioning g-level shall be recorded to determine compliance with the applicable requirements."

Custodian:
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